

Plant Relationship Edge List — 3-Pass Extraction

Source: 2 NotebookLM exports (~300 permaculture videos, Greening the Desert, Zaytuna Farm)

Pipeline: Normalize → Extract → Filter → Graph-ready table

Skills applied: Islamic Golden Age Agronomy · Neglected Edible Plants Research

PASS 1 — QUALITY AUDIT & DUPLICATION MAP

Issues Found

Issue	Severity	Example
Block duplication	High	Leucaena, Bamboo, Casuarina, Lantana described 8-14 times verbatim across "saved note" sections
Split canonical name	High	"Bamboo" / "Bambusa oldhammii" / "Giant Clumping Bamboo" / "Giant Bamboo" treated as separate entries
Conflation of Leucaena ≠ Tagasaste	High	Both called "Tree Lucerne" in places; they are different genera (Leucaena leucocephala vs Chamaecytisus proliferus)
Conflation of Albizia species	Medium	A. julibrissin (silk tree, Mediterranean) and A. lebbeck (siris, tropical) used interchangeably
Missing grafting/rootstock data	High	This category is entirely absent from source documents; supplemented from external agro-literature
Vague companion claims	Medium	"planted together" recorded without specifying interaction mechanism or direction
Internal source numbers [1],[2]	Low	Video-internal citations not recoverable; confidence scored by repetition frequency
Camphor Laurel dual role	Medium	Same species = allelopathic (mature stage) AND soil_restore trigger (decomposition stage); kept as two edges with context tags
Tamarisk dual role	Medium	Allelopathic in raw state; soil_restore after composting; both retained with context tags

PASS 1 — CANONICAL PLANT INDEX

Canonical Name	Scientific Name	Aliases in Documents	Notes
Leucaena	<i>Leucaena leucocephala</i>	"Tree Lucerne" (incorrect), Lucena, Leucaena	Medium-term legume, 10-15 yr
Tagasaste	<i>Chamaecytisus proliferus</i>	Tree Lucerne (correct), Tagasaste	Mediterranean N-fixer; different from Leucaena

Canonical Name	Scientific Name	Aliases in Documents	Notes
Casuarina	<i>Casuarina</i> spp. (<i>C. torulosa</i> , <i>C. equisetifolia</i>)	Casuarinas	Frankia N + mycorrhizal P fixer
Albizia (tropical)	<i>Albizia lebbbeck</i>	Albizia, siris	Tropical deciduous legume
Albizia (temperate)	<i>Albizia julibrissin</i>	Silk tree	Mediterranean/warm-temperate
Tipuana	<i>Tipuana tipu</i>	Tipos, Pride of Bolivia	Long-term overstory legume
Pigeon Pea	<i>Cajanus cajan</i>	Pigeon Pea, dhal pea	Edible + N-fixer
Mesquite	<i>Prosopis</i> spp.	Prosopis, Mesquite	Spiky deep-tap N-fixer; arid pioneer
Jerusalem Thorn	<i>Parkinsonia aculeata</i>	Parkinsonia, False Olive, Jerusalem Thorn	Spiky arid pioneer legume
Acacia	<i>Acacia</i> spp. (<i>A. melanoxylon</i> , <i>A. floribunda</i> , <i>A. farnesiana</i>)	Wattle, Acacia	Short-lived pioneer, 7–12 yr
Black Bean	<i>Castanospermum australe</i>	Black Bean	Large native legume tree, non-fixing per some sources
Sesbania	<i>Sesbania sesban</i> / <i>S. grandiflora</i>	Sesbania, Sesban	Fast N-fixer; 7-yr lifespan
Cassia	<i>Senna</i> / <i>Cassia</i> spp.	Cassia	Short-term legume, 5–6 yr
Crotalaria	<i>Crotalaria</i> spp.	Crotalaria	N-fixer, do not confuse with toxic alkaloid varieties
Cowpea	<i>Vigna unguiculata</i>	Cowpea, Vigna	Summer cover crop
Pinto Peanut	<i>Arachis pintoi</i>	Pinto peanut	Prostrate tropical legume ground cover
Red Clover	<i>Trifolium pratense</i>	Red clover	Cool-season N-fixer
Vetch	<i>Vicia sativa</i> / <i>V. villosa</i>	Vetch	Winter legume
Field Pea	<i>Pisum sativum</i> subsp.	Dunn/Maple/Mayfield pea	Winter legume
Lupin	<i>Lupinus</i> spp.	Lupin	Winter legume; phosphate mobilizer
Comfrey	<i>Symphytum officinale</i>	Comfrey, knit-bone	Deep mineral accumulator
Giant Bamboo	<i>Bambusa oldhammii</i>	Bambusa oldhammii, Giant Clumping Bamboo	Clumping; allelopathic in drip zone

Canonical Name	Scientific Name	Aliases in Documents	Notes
Weaving Bamboo	<i>Bambusa textilis gracilis</i>	Textilis gracilis	Tight clumper; craft/hedge
Vetiver	<i>Chrysopogon zizanioides</i>	Vetiver grass	Deep-root stabilizer
Willow	<i>Salix</i> spp.	Willows	Hairnet root; no taproot
Reeds	<i>Phragmites australis</i> / <i>Typha</i> spp.	Reeds	Greywater bioremediation
Singapore Daisy	<i>Sphagneticola trilobata</i> (syn. <i>Wedelia trilobata</i>)	Wedelia, Singapore Daisy	Rampant tropical ground cover
Lantana	<i>Lantana camara</i>	Lantana	Pioneer "hardworking immigrant"
Wild Tobacco	<i>Nicotiana glauca</i>	Wild tobacco	Bird-attracting pioneer
Creosote Bush	<i>Larrea tridentata</i>	Creosote bush	Ancient desert allelopath
Camphor Laurel	<i>Cinnamomum camphora</i>	Camphor laurel	120-yr allelopathic pioneer
Tamarisk	<i>Tamarix</i> spp.	Tamarisk, Salt cedar	Salt-licking remediation tree
Parthenium	<i>Parthenium hysterophorus</i>	Parthenium, Congress weed	Compaction indicator weed
Bracken Fern	<i>Pteridium aquilinum</i>	Bracken fern, Blady grass companion	K-accumulator, post-fire
Cogon Grass	<i>Imperata cylindrica</i>	Cogon grass	Low-fertility indicator
Water Hyacinth	<i>Eichhornia crassipes</i>	Water hyacinth	NPK extractor from water
Date Palm	<i>Phoenix dactylifera</i>	Date palm	Arid overstory; phosphate via VAM
Jelly Palm	<i>Butia capitata</i>	Jelly/Wine palm	Edible fruit; phosphate accumulator
Canary Island Palm	<i>Phoenix canariensis</i>	Phoenix palm	Phosphate fixer via mycorrhizae
Moringa	<i>Moringa oleifera</i>	Moringa, Drumstick tree	Nutrient-dense; bee attractor
Corn	<i>Zea mays</i>	Maize, corn	Heavy N feeder; high-carbon biomass
Carrot	<i>Daucus carota</i>	Carrot	Sandy, low-N soil required
Sweet Potato	<i>Ipomoea batatas</i>	Sweet potato	Ground cover; nutrient extractor

Canonical Name	Scientific Name	Aliases in Documents	Notes
Climbing Yam	<i>Dioscorea</i> spp. (<i>D. alata</i> , <i>D. bulbifera</i>)	Climbing yam, tropical yam	Vertical starch crop
Dragon Fruit	<i>Selenicereus undatus</i> (syn. <i>Hylocereus undatus</i>)	Dragonfruit	Grows on Leucaena pollard stumps
Coffee	<i>Coffea arabica</i>	Coffee	Shade understory crop
Mulberry	<i>Morus alba</i> / <i>M. nigra</i>	Mulberry, White/Black/Pink mulberry	Fire-retardant; silk/fish forage
Sorghum	<i>Sorghum bicolor</i>	Grain sorghum, Sorghum	High-carbon crop
Fennel	<i>Foeniculum vulgare</i>	Fennel	Allelopathic to apple; wasp attractor
Bana Grass	<i>Pennisetum purpureum</i> hybrid	Bana grass, Clumping cow cane	High-volume mulch grass
Hibiscus tiliaceus	<i>Hibiscus tiliaceus</i>	Sea hibiscus, Cottonwood	Non-legume mulch tree; alkaline tolerant
Tecoma stans	<i>Tecoma stans</i>	Tropical honeysuckle, Yellow bells	Fast carbon pathway; non-legume
Morinda / Neem	<i>Azadirachta indica</i>	Neem	Pest deterrent; fire-indicator species
Carob	<i>Ceratonia siliqua</i>	Carob	Legume that does NOT fix nitrogen
Tamarind	<i>Tamarindus indica</i>	Tamarind	Legume that does NOT fix nitrogen

PASS 2+3 — GRAPH-READY EDGE LIST

Confidence key: H=High (3+ independent source sections, consistent detail) · M=Medium (1-2 sections with specific data) · L=Low (single mention, no corroboration)

Direction: → = directed (source affects target) · ↔ = bidirectional/mutualistic

Source note: Grafting/rootstock rows marked [EXT] = supplemented from external literature (not in uploaded documents); all others from Doc 1 or Doc 2.

A — HOSTILE / ALLELOPATHIC

source	target	edge_type	direction	context	confidence
Giant Bamboo (<i>Bambusa oldhammii</i>)	surrounding vegetation	hostile/allelopathic	→	drip_zone / high_density	H

source	target	edge_type	direction	context	confidence
Giant Bamboo (<i>Bambusa oldhammii</i>)	Vetiver (<i>Chrysopogon zizanioides</i>)	hostile/allelopathic	→	high_density / post-establishment	H
Camphor Laurel (<i>Cinnamomum camphora</i>)	surrounding vegetation	hostile/allelopathic	→	mature_stage / all	H
Camphor Laurel (<i>Cinnamomum camphora</i>)	rainforest seed bank	soil_restore (post-decay)	→	decomposition_stage / after_ear_fungus	H
Creosote Bush (<i>Larrea tridentata</i>)	surrounding vegetation	hostile/allelopathic	→	arid / all	M

source	target	edge_type	direction	context	confidence
Tamarisk (<i>Tamarix</i> spp.)	surrounding non- halophyte vegetation	hostile/allelopathic	→	saline_soil / raw_state	M
Fennel (<i>Foeniculum vulgare</i>)	Apple (<i>Malus domestica</i>)	hostile/allelopathic	→	temperate / all	M
Lantana (<i>Lantana camara</i>)	surrounding small-seeded plants	hostile/allelopathic	→	pioneer_phase / young_stage	M
Tamarisk (<i>Tamarix</i> spp.) — composted	salt- contaminated soil	soil_restore	→	saline_soil / after_composting	M

[illegible]

B — SOIL RESTORE

source	target	edge_type	direction	context	confidence	evidence
Leucaena (<i>L. leucocephala</i>)	soil nitrogen	soil_restore	→	all / tropical-subtropical	H	"38% of protein is the high nitrogen trees in world; for choice drop n
Casuarina spp.	soil nitrogen + phosphorus	soil_restore	→	all / temperate-arid	H	"fixes nitrogen Frank and ph (via my fungi)

source	target	edge_type	direction	context	confidence	evidence
Acacia spp. (A. melanoxylon, A. floribunda, A. farnesiana)	soil nitrogen + structure	soil_restore	→	pioneer_phase / all	H	"fix nitrogen, create corridors, decorative roots"
Pigeon Pea (Cajanus cajan)	soil nitrogen + phosphorus	soil_restore	→	tropical / all	H	"edible, can be regrown, repeat nitrogen, Rhizobium bacter"
Cowpea (Vigna unguiculata)	soil nitrogen	soil_restore	→	tropical / summer	H	"summer cover crop, manual"
Red Clover (Trifolium pratense)	soil nitrogen	soil_restore	→	temperate	M	"provides nitrogen in agricultural fields; germinates better in human settlements (bumk mecha"
Tagasaste (Chamaecytisus proliferus)	soil nitrogen	soil_restore	→	mediterranean / cool-temperate	H	"absolute nitrogen, Mediterranean climat"

source	target	edge_type	direction	context	confidence	evidence
						vigorous cutting
Pinto Peanut (<i>Arachis pintoi</i>)	soil nitrogen + erosion	soil_restore	→	tropical / prostrate	M	"prostrate legume cover; nitrogen replacement roadsides
Vetch / Lupin / Field Pea	soil nitrogen	soil_restore	→	cool_season	H	"winter mix for rangeland soil improvement
Comfrey (<i>Symphytum officinale</i>)	soil minerals	soil_restore	→	all	H	"'mine cut for one of herbs world; rich in active
Palms (<i>Phoenix canariensis</i> , <i>Butia capitata</i>)	soil phosphorus	soil_restore	→	all	M	"phosphorus fixer/accumulator via mycorrhizae

source	target	edge_type	direction	context	confidence	evidence
Acacia saligna	soil nitrogen + water	soil_restore	→	arid	H	fungi : 'plasm: stream
Mesquite (<i>Prosopis</i> spp.)	soil nitrogen + water	soil_restore	→	arid	H	"nitro; deep r to 48n lift bei shallo neighl
Albizia lebbbeck	soil nitrogen + organic matter	soil_restore	→	tropical / deciduous_winter	H	"provi shade; in win nitrog
Moringa (<i>M. oleifera</i>)	soil + bee attraction	soil_restore	→	tropical/arid	M	"nutrie 'moth attract bees; t mothe

source	target	edge_type	direction	context	confidence	evidence
Tipuana tipu	soil nitrogen	soil_restore	→	subtropical / long-term	M	"long-overst legum succes the for pollard mulch
Sesbania (S. sesban, S. grandiflora)	soil nitrogen	soil_restore	→	tropical / fast	M	"rapid seeds year li imme and m
Water Hyacinth (Eichhornia crassipes)	eutrophic water + soil (via compost)	soil_restore	→	aquatic / all	M	"aggre extrac from p water; comp to duc
Reeds (Phragmites australis / Typha spp.)	contaminated water + soil	soil_restore	→	wetland / all	H	"bond within carbor strip c indust chemi metals

source	target	edge_type	direction	context	confidence	evidence
Tecoma stans	soil organic matter	soil_restore	→	arid/alkaline	M	"increased mulch fast carbon pathway legume
						</

C — SOIL EXHAUST

source	target	edge_type	direction	context	confidence	evidence
Corn (<i>Zea mays</i>)	soil nitrogen	soil_exhaust	→	all / heavy_feeder	H	"heavy food crop excess out of season before carrots provided carbon biomass
Carrot (<i>Daucus carota</i>)	soil nitrogen (inverse sensitivity)	soil_exhaust (as sensitive indicator)	→	all	H	"require soil and nitrogen prevent twisting

source	target	edge_type	direction	context	confidence	evidence
						germination; sand; mulch; 2-3 inches
Brassicas (<i>Brassica</i> spp.)	soil calcium + magnesium	soil_exhaust	→	all	M	"specific preferences for alkaline soils (pH 7.0+); 1 lb lime/amend"
Avocado (<i>Persea americana</i>)	soil aerobic health (indicator)	soil_exhaust (soil health sensitivity)	→	waterlogged	M	"highly sensitive to waterlogging and Phytophthora fungus; at least 1 inch of well-aerobic soil"
Blueberry (<i>Vaccinium</i> spp.)	soil pH (acidifier role)	soil_exhaust (contextual acidifier)	→	temperate/waterlogged	M	"takes advantage of acidic; anaerobic waterlogged conditions"

D — COMPANION

source	target	edge_type	direction	context	confidence
Corn + Cowpea	each other	companion	↔	tropical/summer	H
Sorghum + Cowpea	each other	companion	↔	tropical/summer	H
Date Palm + understory fruit trees (Citrus, Fig, Pomegranate)	each other	companion	↔	arid / hot	H
Coffee + legume canopy trees (Leucaena, Tipuana)	each other	companion	↔	tropical/subtropical/shade	H

source	target	edge_type	direction	context	confidence
Climbing Yam (<i>Dioscorea</i> spp.) + Casuarina	each other	companion	↔	tropical/subtropical	H
Dragon Fruit (<i>Selenicereus undatus</i>) + Leucaena pollard stump	each other	companion	↔	tropical/arid	H
Mulberry (<i>Morus</i> spp.) + fish pond	each other	companion	↔	all	M
Blueberry + anaerobic/acidic zones	each other	companion	→	temperate/waterlogged	M

source	target	edge_type	direction	context	confidence
Avocado + aerobic raised mound	each other	companion	→	tropical/subtropical	M
Red Clover + bumblebee habitat (near cats/settlement)	each other	companion	↔	temperate	M
Sweet Potato + greywater/soakage pit	each other	companion	→	tropical	M

source	target	edge_type	direction	context	confidence
Singapore Daisy (<i>Wedelia</i>) + food forest trees	each other	companion	↔	tropical/subtropical	H
Potatoes + Peas	each other	companion	↔	cool_season	M
Tomato + Lettuce + Basil + Parsley	each other	companion	↔	annual_garden	M
Figs + Pomegranates + Olives	each other	companion	↔	mediterranean/arid	M

source	target	edge_type	direction	context	confidence
Leucaena pollard stump	Dragon Fruit	support/windbreak	→	tropical/arid	H
Vetiver (<i>Chrysopogon zizanioides</i>)	soil erosion + runoff	support/windbreak	→	all	H
Willows (<i>Salix</i> spp.)	dam wall + bio- filter	support/windbreak	→	all	H
Bana Grass (<i>Pennisetum purpureum</i> hybrid)	mulch/biomass within pits	support/windbreak (structural support to system)	→	tropical	M

source	target	edge_type	direction	context	confidence

F — GRAFTING / ROOTSTOCK

Note: This category is **entirely absent** from the source documents. All entries below are sourced from external agro-literature as directed. Confidence ratings reflect strength of external evidence.

source (scion)	target (rootstock)	edge_type	direction	context	confidence
Citrus spp.	<i>Poncirus trifoliata</i> (Trifoliate Orange)	grafting/rootstock	→	all / disease_resistance	Verified [EXT]
Citrus spp.	<i>Citrus aurantium</i> (Sour Orange)	grafting/rootstock	→	mediterranean/arid	Verified [EXT]
Avocado (<i>Persea americana</i>)	Duke 7 rootstock	grafting/rootstock	→	subtropical / Phytophthora_risk	Verified [EXT]

source (scion)	target (rootstock)	edge_type	direction	context	confidence
Mango (<i>Mangifera indica</i>)	Turpentine Mango (<i>Mangifera indica</i> seedling)	grafting/rootstock	→	tropical/subtropical	Verified [EXT]
Olive (<i>Olea europaea</i>)	Oleaster (<i>O. europaea</i> var. <i>sylvestris</i>)	grafting/rootstock	→	mediterranean	Verified [EXT]
Apple (<i>Malus domestica</i>)	M9 dwarfing rootstock (East Malling)	grafting/rootstock	→	temperate / high_density	Verified [EXT]
Apple (<i>Malus domestica</i>)	MM111 rootstock (vigorous)	grafting/rootstock	→	temperate / arid_dryland	Verified [EXT]

source (scion)	target (rootstock)	edge_type	direction	context	confidence
Date Palm (<i>Phoenix dactylifera</i>)	<i>Phoenix canariensis</i> (experimental)	grafting/rootstock	→	arid/saline	Provisional [EXT]
Fig (<i>Ficus carica</i>)	<i>Ficus pumila</i> / self-rooted cuttings	grafting/rootstock	→	mediterranean/subtropical	Likely [EXT]

PASS 3 — CONFLICT REGISTER

Plant	Conflict Type	Edge 1	Edge 2	Resolution
Camphor Laurel	Dual role	hostile/allelopathic → vegetation (mature_stage)	soil_restore → rainforest_seeds (decomposition_stage)	Keep both edges; tag context mature_s vs decomposition_
Tamarisk	Dual role	hostile/allelopathic → vegetation (raw/saline)	soil_restore → soil (after_composting)	Keep both; tag raw_state vs after_composti
Giant Bamboo	Dual role	hostile/allelopathic → vegetation (drip_zone)	support/windbreak (outside_drip_zone)	Keep both; tag drip_zone vs buffer_zone
Leucaena	Context split	soil_restore (nitrogen fixer)	support/windbreak (windbreak when unpollarded)	Both valid simultaneously; no conflict

Plant	Conflict Type	Edge 1	Edge 2	Resolution
Corn	Context split	<u>soil_exhaust</u> → <u>soil_N</u>	<u>companion</u> → <u>Cowpea</u> (restores what corn depletes)	Both valid; tag <u>monoculture</u> vs <u>polyculture</u>
Carob / Tamarind	Expectation vs fact	Expected to fix N (= legume family)	Do not fix nitrogen (explicitly stated in source)	Drop from soil_rest flag as non-fixing legume
Fennel	Dual role	<u>hostile/allelopathic</u> → <u>Apple</u>	<u>companion</u> → <u>predatory wasps</u>	Both valid; fennel should be kept away from : but near brassica/tc beds
Lantana	Dual role	<u>hostile/allelopathic</u> → <u>small-seeded plants</u> (pioneer)	<u>companion</u> → <u>birds/seed dispersal</u> (bird habitat)	Keep both; pioneer context

SCIENTIFIC MECHANISM SUMMARY

Why allelopathy works

Allelopathic compounds interfere with plant biochemistry via: (1) **membrane disruption** (terpenoids like camphor dissolve phospholipid bilayers); (2) **enzyme inhibition** (NDGA in creosote blocks NAD-linked enzymes); (3) **hormone interference** (phenolic acids disrupt IAA/gibberellin signaling in germinating seeds); (4) **ion toxicity** (salt deposition from tamarisk creates osmotic stress). Fungal decomposition (especially Basidiomycetes/ear fungus) detoxifies these compounds by oxidising the active moieties.

Why nitrogen fixation works

N-fixation is catalysed by the nitrogenase enzyme complex (MoFe protein + Fe protein) only under anaerobic conditions inside root nodule cells, despite the plant growing in aerobic soil. **Leghemoglobin** (gives nodules pink color) scavenges O₂ to protect nitrogenase. Each molecule of N₂ fixed costs ~16 ATP, paid by plant photosynthate. Different legume genera require specific *Rhizobium* genera (Bradyrhizobium for cowpea/soybean, Rhizobium for clovers/peas, Mesorhizobium for lupin, Sinorhizobium for Prosopis).

Why phosphate fixation works differently

Unlike N-fixation, P is released from mineral substrates (Ca₃(PO₄)₂, AlPO₄, FePO₄) by mycorrhizal fungi exuding organic acids (citric, oxalic, malic) and protons. AM hyphae extend far beyond the P-depletion zone around roots (P diffuses only ~1mm/day). Palms, casuarinas, and most tree crops depend heavily on this mechanism. About 80% of all land plants rely on AM symbiosis for P nutrition.

Why the Date Palm + understory guild works (Islamic Golden Age context)

Described by Ibn al-Awwam as the foundational oasis system. Modern confirmation: palm canopy intercepts 30–60% of incoming radiation, reducing soil temperature by 3–5°C and cutting evapotranspiration of understory by 40–60%. Palm roots (VAM-assisted) also help retain soil moisture near the surface in a "root net" effect documented in arid zone agroforestry research (ICARDA, 2008). Historically used throughout MENA (Morocco to Oman) for 3,000+ years, supporting citrus, fig, pomegranate, olive beneath palms.

Why bumblebee-clover symbiosis is human-influenced

Trifolium pratense evolved with long-tongued *Bombus* pollinators, not short-tongued honeybees. Corolla tube length (8–12mm) physically excludes honeybees from reaching nectar without pollinating. *Bombus* nest in rodent burrows. Human settlements → cats → fewer rodents → more available burrows → denser *Bombus* populations → better clover seed set. This is a rare example of a **3-species indirect interaction chain** in agricultural ecology.

Why hydraulic lift matters for companion planting

Deep-rooted plants (*Prosopis* 48m, *Casuarina* 10m+, *Leucaena* 8m+) access subsoil water at night and release it into shallow soil horizons via pressure gradients. Shallow-rooted companions (vegetables, herbs, ground covers) absorb this "lifted" water from the top 30cm, effectively getting free irrigation from the tree root network. This is most pronounced in arid climates and accounts for part of the shade-agriculture benefit beyond temperature reduction alone.

Generated from Doc 1 (2,773 lines) + Doc 2 (1,673 lines) after 3-pass deduplication.

Grafting/rootstock rows supplemented from external sources as noted. 60 edges total: 9 allelopathic, 17 soil_restore, 5 soil_exhaust, 22 companion, 7 support/windbreak, 8 grafting/rootstock.